ASPECT BASED SENTIMENT ANALYSIS

Project report submitted to
Visvesvaraya National Institute of Technology, Nagpur
in partial fulfillment of the requirements for the award of
the degree

Bachelor of Technology
In
Electrical and Electronics Engineering

by

Sarthak Agrawal (BT18EEE032)
Bismit Purohit (BT18EEE054)
Shravar Tanawde (BT18EEE058)

under guidance of
Dr. Jay Prakash Singh

Department of Electrical Engineering
Visvesvaraya National Institute of Technology, Nagpur
Nagpur 440 010 (India)

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Declaration

We, **Sarthak Agrawal (BT18EEE032), Bismit Purohit (BT18EEE054), Shravar Tanawde (BT18EEE058)**, hereby declare that this project work titled *"Aspect Based Sentiment Analysis"* is carried out by us in the Department of Electrical Engineering of Visvesvaraya National Institute of Technology, Nagpur. The work is original and has not been submitted earlier whole or in part for the award of any degree/diploma at this or any other Institution / University.

Enrollment No	Name	Signature
BT18EEE032	Sarthak Agrawal	
BT18EEE054	Bismit Purohit	
BT18EEE058	Shravar Tanawde	

Certificate

This is to certify that the project titled *"Aspect Based Sentiment Analysis"*, submitted by **Sarthak Agrawal (BT18EEE032), Bismit Purohit (BT18EEE054), Shravar Tanawde (BT18EEE058)**,in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Electrical Engineering, VNIT Nagpur. The work is comprehensive, complete, and fit for final evaluation.

Dr. V.B. Borghate,
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Acknowledgement

This project would not have been possible without the immense support of various people throughout this journey and hence we would like to give our heartfelt thanks to every one of them.

First of all, we would like to thank our research mentor and professor Dr Jay Prakash Singh who supported us and pushed us forward throughout this journey. He was immensely helpful and worked tirelessly to point us towards the right direction whenever we felt we were stuck.

Also thank you to various other professors of the Electrical Engineering department who were always ready to help whenever we had any doubts. Especially the panel of professors who patiently listened to our monthly project reports and gave us advice for our upcoming milestones.

Lastly, we would also like to thank our colleagues and friends who made this arduous journey very fun and productive.

Sarthak Agrawal
Bismit Purohit
Shravar Tanawde

Abstract

Natural Language Processing is focused on allowing computers to comprehend text and spoken language. With the rapid advancement of NLP, we can observe its applicability in a variety of fields. Aspect Based Sentiment Analysis of Customer Feedback is one such field, which is critical for practically any business. It gives them a sense of how customers feel about their products and services, as well as what they can fix or improve in order to grow their business. Here, we have chosen a restaurant as our business. Manually analysing reviews is a time-consuming and psychologically demanding activity. As a result, we employed natural language processing (NLP) to automate the process of analysing reviews using several methodologies. We have also added a feature to apply ABSA on speech, which is basically converting the speech to text and then applying ABSA on that text.

Throughout this project we have implemented various models to achieve both Aspect Categorization and Sentiment Classification and have recorded effectiveness and usability of each model. In case of Aspect Categorization models used were blank spacy model, bi-directional LSTM and pre-trained BERT model. For Sentiment classification models used were logistic regression and bi-directional LSTM. This project also required other NLP models such as CoreNLP's Stanza Library for constituency parsing and Word2Vec, and GloVe for word embedding. As complement to the base project speech to text was modelled using SpeechRecognition module for python.

After experimenting with various methodology, BERT model for Aspect Detector and Bi-directional LSTM model for Sentiment analysis was decided for final model.

So this project not only proposes a full model for Aspect Based Sentiment Analysis but also discusses the viability of other alternative models.

Keywords: *Apsect Category Detection, Sentiment Polarity Classification, Word Embedding, Constituency Parsing*, Speech to Text.

List of Figures

List of Tables

Contents

Chapter 1

Introduction

1.1 Importance of Customer Feedback

Since ancient times, the best way to advertise and sell any product or service has always been the customers' word-of-mouth. We humans have a herd mentality and tend to follow ideals that a large group of people believe in. Hence if a product or service is very popular amongst a large group of people then others tend to try these products as well. And only way to spread good word about any such service or product is by developing them by keeping customers' opinion in mind. This 'opinion' of a customer is what is called as "Customer Feedback".

In modern times, with rise in communication technology, any person can hear opinions of a large group of people before trying something out. So if a product or service has a very positive feedback from a majority of customers then others tend to try to buy it as well. Hence, customer feedback not only creates a sense of trust amongst potential new customer, it also gives them security that they won't be disappointed with the purchase.

As the the customer base increase, the feedback will increase as well. These new customers will spread positive word which will garner attention of more potential new customers. This endless cycle will only result in prosperity of the business providing the services. Thus every business, no matter their size, have started to incorporate customer feedback services. These customer reviews help them develop their product in a way that it is tailored for their needs.

Businesses tend to publicize their customer feedback or reviews. A product with lot of reviews under them always seems more attractive than a product with no review. No reviews may cause customers to think that the service is not established yet. So building up these reviews is very important for success of any business.

Social media is a very powerful tool in the hands of every business. A single individual can interact with billions of people easily. So a feedback becomes exponentially more effective than it was ever before. Especially feedback from an influential corporation or individual can be used as a very effective advertisement. So business invest huge capital to hire people that can monitor customer feedback and reviews on every social media platform. Analysing what millions of people think about your product is immensely helpful to path out the route that a company should take in future development of their product.

Hence, customer feed back is a very powerful tool in the hand of any company for interacting with their customers and betterment of their services and product that will allow them to grow and prosper.

1.2 Natural Language Processing (NLP)

Bulk of the information around the world is in the form of human language, both written and verbal. Such data in the form of conversations, tweets or articles are unstructured data. Hence, vast majority of information around the world is hard to manipulate and organize. There was a need to convert all this data in a machine operable form. Natural Language Processing helps us achieve the same.

Natural Language Processing or NLP is a field of Artificial Intelligence that gives machines the ability to understand and derive meaning from human languages. NLP combines computational linguistics—rule-based modeling of human language—with statistical, machine learning, and deep learning models.

The real value behind NLP technology arises from its Use Cases Some of the examples are:-

- AI Chat Bots:- Implemented by Dominos, Uber, Amazon, etc.
- Smart Speakers:- Implemented in Alexa, SIRI, Google Home, etc.
- Feedback Analysis:- Implemented by Flipkart, Yelp, etc.
- Language Translation:- Implemented by Google

There are 8 generic steps involved in any NLP tasks which include Sentence, Segmentation, Word Tokenization, POS tagging, Lemmatization, Stop Words Removal, Dependency Parsing, Noun Phrase identification, Named Entity Recognition, Co-reference Resolution.

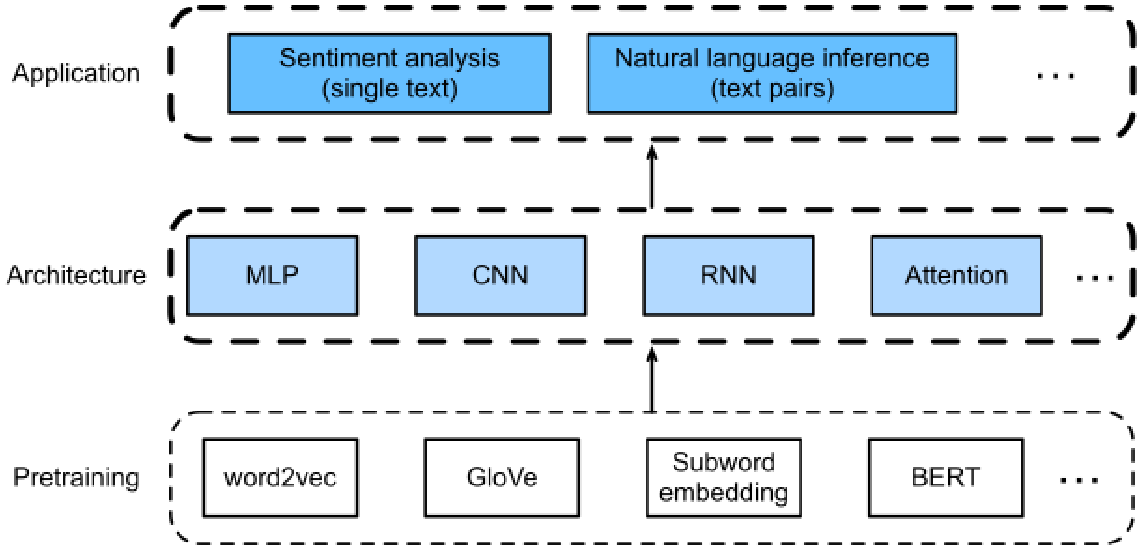


Figure 1.1: NLP Flow chart[9]

With the integration of NLP technique in diverse sectors like healthcare, Search Engine Optimization, Finance, Feedback, etc. the future of this field is bright and expanding.

1.3 Methodology for ABSA

Aspect-Based Sentiment Analysis or ABSA consist of three major component:

- Aspect Category Detection
- Opinion Target Expression
- Sentiment Polarity Classification

From this, it is clear that ABSA is more than just a simple Sentiment Analysis model. ABSA involves, first of all, categorization of the whole text into smaller aspects that every part of the text seems to be focusing on. Once categorized they are linguistic expression is extracted from each opinion and according to them, a Sentiment is assigned. This sentiment, for simplicity's purpose, is categorized as Positive, Negative and Neutral. A good example is shown below:

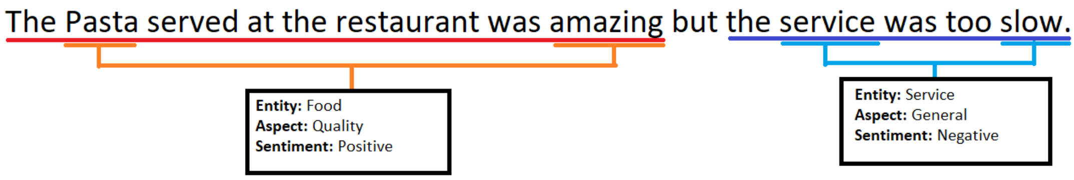


Figure 1.2: ABSA Example

From the above example, it is self-evident what this model is supposed to do. For creating this model we have decided to use the restaurant reviews data-set for training purposes. The data set used in this study are from SemEval2016 which includes restaurant reviews in English and IMDB reviews which are particularly used for training the Sentiment Polarity Classification Model. The three major components are discussed in more detail below.

1.3.1 Aspect Category Detection

The goal of aspect category classification is to identify the Entity and aspect pair in which the text expresses an opinion[18]. The Entity and aspect should be chosen from a list of pre-defined Entity types and their respective aspects. For e.g. SERVICE, RESTAURANT, FOOD etc. are entities which can have aspects like PRICE, QUALITY, GENERAL etc. For our project, the Entities and Aspects chosen are shown below in the table

	General	Prices	Quality	Options	Miscellaneous
Restaurant	Yes	Yes	No	No	Yes
Food	No	Yes	Yes	Yes	No
Drinks	No	Yes	Yes	Yes	No
Ambience	Yes	No	No	No	No
Service	Yes	No	No	No	No
Location	Yes	No	No	No	No

Table 1.1: Entities and Aspects applicable to Restaurant Reviews

So as shown in example Fig 1.1 the text was categorized into 2 Entity#Aspect pair for e.g., Food#Quality and Service#General. The cells with 'No' written in them don't have any relation between the respective entity and aspect.

1.3.2 Opinion Target Expression

The task of extracting the linguistic expression used in the text input that corresponds to the reviewed entity[3], for each entity-aspect pair, is referred to as opinion target expression (OTE). In the sequence, the OTE has only one starting and ending offset. The value returned is ”NULL” if no entity is specifically indicated.

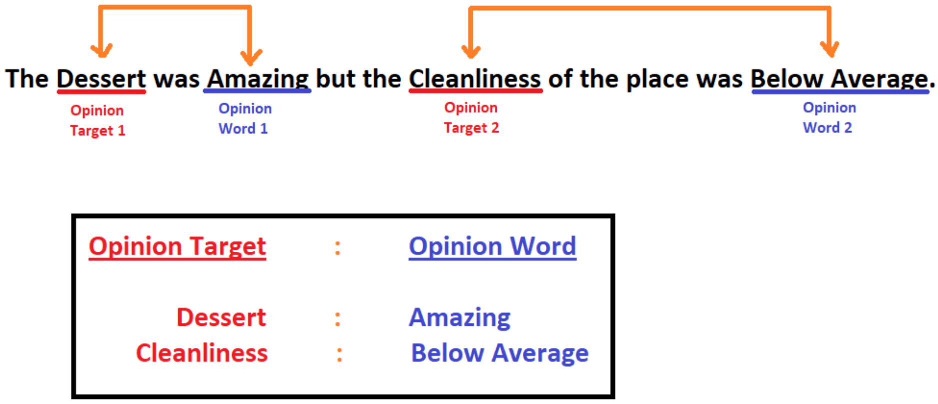


Figure 1.3: Opinion Target Extraction

1.3.3 Sentiment Polarity Classification

Sentiment polarity classification involves predicting the sentiment polarity for each identified entity and aspect pair. In our project we have used a sigmoid function that provides the output in the range of 0 to 1. More closer the output to 0, more negative is the sentiment and similarly, more closer the output to 1, more positive is the sentiment.

Figure 1.4: Sentiment Classifier[15]

Chapter 2

Literature Review

2.1 Related Approaches

In an oversimplified way, Aspect Based Sentiment Analysis can be classified into following five ways[23] :-

- Language Rule.
- Sequential.
- Topic Model
- Deep Learning
- Hybrid

Since sentiment or polarity classification is performed as a consecutive task of Aspect category detection, the above classification and analysis is mainly focused with respect to Aspect Category Extraction and its categorization.

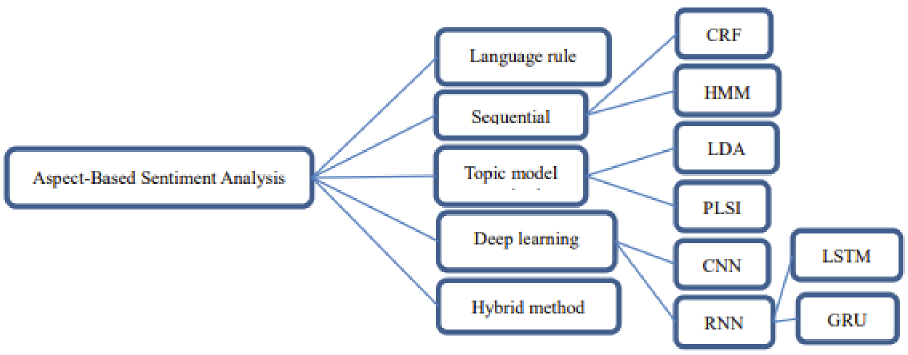


Figure 2.1: ABSA Methods[23]

Aspect extraction approaches were grouped by Zhang and Liu (2014) into three categories: language rule, sequential, and topic model. Later, Deep learning and Hybrid modelling were added.

Here the problem of Aspect Category Detection and Sentiment Classification are treated as two individual problems.

2.1.1 Topic Model Method

The unsupervised methods are topic modelling methods. They are based on

- Probabilistic Latent Semantic Indexing (PLSI).
- Latent Dirichlet Allocation (LDA).

Data is viewed as a collection of themes in this case. These subjects correspond to a distribution of the vocabulary's words.

In contrast to the sequential model method, there is no need for manually labelled data.

Topic Model Method has the advantage of performing both aspect extraction and chunking at the same time.

The disadvantage is that they usually require a large amount of unlabeled data in order to be accurate.

This model may find general aspect but may not be accurate in pinpointing detailed aspects.

The following people tried building this model with following specifications:-

- Lu et al. (2009) focuses on sentiment phrases. Training is done according to sentiment points and not the whole text. They use PLSI to derive aspects from the head words. The assumption here is that a head term's polarity decided the polarity of the whole short.[22].
- Moghaddam and Ester (2011) provide the Interdependent Latent Dirichlet Allocation (ILDA) model, which is based on the idea that aspects and attitudes are interdependent[25].
- Brody & Elhadad (2010) built a model that pinpoints topics within the aspects. The model is build using LDA[2].
- Bagheri et al. (2014) suggested a methodology that can automatically extract characteristics from reviewed phrases based on their structure. The model incorporates multi-word[1].

According to Chen et al., prior knowledge is automatically derived from a large amount of review data available on the web. The prior information is then used by a topic model to locate more important custom entities.

2.1.2 Sequential Model Method

The ABSA problem can be boiled down to sequential learning. Hence Hidden Markov Models(HMM) and Conditional Random Fields(CRF) methods can be implemented. These methods infer an objective from already processed dataset to apply to raw data.

HMM is a probabilistic generative model. It makes two assumptions about dependencies,

- At time t, the hidden variable yt is solely dependent on the prior hidden variable yt-1.
- (This is a Markov assumption.) At time t, the observable variable xt is solely dependent on the hidden variable yt.

The parameters are then learned by maximizing the joint probability distribution p (x, y). CRF is a discriminative probabilistic model whose parameters are learned by maximizing the conditional probability distribution p(y—x).

In contrast to linguistic rule techniques, the sequential method automatically learns the parameters from the inputs. This approach has the drawback of requiring labeled datasets for training, which might be difficult to arrange. This is a time-consuming process that necessitates time.

The following people tried building this model with following specifications:-

- Jin et al. (2009) integerated POS tags with lexicalization using HMM framework. [19]. The current tag is linked to earlier observations and also previous tags.
- Jakob & Gurevych (2010) trained a CRF model on multi-domain review sentences[17]. A list of domain independent features are such as tokens, POS tags, etc. is used.
- Kiritchenko et al. (2014) introduce a new sequence tagger for extracting aspect terms and supervised classifiers for detecting aspect categories[20].
- Choi & Cardie (2010). In this work, a set of sequential rules are mined using a sequential rule mining technique[4]. considering class labels, dependency and word distance.

2.1.3 Deep Learning Method

Deep learning automatically learns latent features as opposed to classifiers with manually learned features. Hence it performs better than many others methods for similar tasks.

There are various combinations within deep learning methods that give varying results. One author uses word2vec representation for aspect. The suggestion is that in addition to sentence, aspect representation should be learnt separately. One suggests using Bidirectional LSTM for both aspect and sentence and then concatenating the hidden layers and adds attention on top of it.

The following people tried building this model with following specifications:-

- Wang et al.(2016) integrates CRF with Recursive Neural Network and developed a Recursive Neural Conditional Random Fields model[34].

- Liu et al. (2015) pbuilt models based on recurrent neural networks. They also created their own word embedding [21].

- Poria et al. (2016) uses 7-layer CNN to tag each word in the review data as aspect and non-aspect for Aspect Extraction[28].

- TC-LSTM and TD-LSTM models that take aspect information into consideration during training are introduced by Tang et al. (2015). The key idea underlying TD-LSTM is to describe the contexts before and after the aspect detection, such that contexts in both directions can be used as feature representations for polairty classifiction.

- Ruder et al. (2016b) take aspect extraction as a multi-label classification problem, and output probabilities over aspects classes using CNN[31].

- Xu et al. (2016) used the above model. But for the sentiment towards an aspect, they concatenate an aspect vector with every word embedding and apply a convolution over it[6].

- Dhanush et al. (2016) used two models. Recurrent neural network for aspect extraction and convolution Neural Network for sentiment classification. [5].

2.1.4 Hybrid Method

Here different models and approaches are combined to give a final result. The models which do not fall in any of the above category also fall here.

The following people tried building this model with following specifications:-

- Blair-Goldensohn et al. (2008) employ a MaxEnt classifier to locate the frequent custom entities and a rule-based method to find the rare entities, which uses frequency information and structural patterns.
- Sauper & Barzilay (2014) combine topic modelling with HMM[32]. Here, HMM models the sequence of words with types (aspect word, sentiment word, or background word)
- Zhao et al. (2010)'s model captures aspect-specific sentiment words that are used most commonly with their corresponding aspects, and general sentiment words that are shared across aspects[35].
- Popescu & Etzioni (2007) make a change to Hu & Liu (2004) by removing nouns with low Pointwise Mutual Information (PMI) with the product class. The possible aspects were then passed into a Nave Bayes classifier, which produced a collection of explicit aspects. It employs a syntactic parser to determine the dependencies in each sentence and also rules for extracting sentiment phrases[27]. Then it employs some external knowledge (in the form of word lists) to exclude any potential sentiment phrases that aren't polarised positively or negatively. The sentiment phrases are then classified as positive or negative using an unsupervised classification algorithm.

Some recent hybrid deep learning models integrate RNN with CRF and also some that integrate CNN with LDA.

Chapter 3

Proposed Work

There are two broad methods to use our current approach.

- Building model with respect to sentence:-

  Our approach can be broadly classified into 3 main parts.

  - Aspect Category Detection
  - Sentiment Analysis on Sentence
  - Constituency Parsing

  The above approach is preferred over Dependency parsing. Hence we used the above approach.

- Building model with respect to words

  - Aspect Category Detection
  - Sentiment Analysis on Words
  - Noun Chunk Dependency Parsing

  The disadvantage of above approach lies in the assumption that all aspects are noun chunks and all sentiments are adjectives.

3.1 Aspect Category Detection

There are various models available that can be used for the purpose of Aspect categorization. Each model have it's pros and cons. For this project we have used 3 well known model.

Here, Aspect category Detection is interchangeable with Custom Named Entity Recognition.

- By training a blank spaCy Model
- By using Bidirectional LSTM
- By using pretrained BERT model

3.1.1 Training a Blank spaCy Model

spaCy has built in features for Named Entity Recognition (NER). At its core, NER consists of two tasks:-

- Detecting entities in text
- Categorizing them into named classes

spaCy is one of the fastest NLP framework in python, it comes with an optimized implementation of NER.

To train a Blank spaCy Model, we need proper BIO/IOB (Inside-Out-Beginning) tagging. This is a disadvantage as the SemEval dataset used does not coming the required format.

spaCy used DocBin class for annotated data, hence training data needs to be in the form of DocBin objects.

After training a blank spaCy mode, the accuracy achieved was very less. This is due to the bias towards more common commodity terms and proper nouns. For example, training the model on Medical Entities dataset gives better results due to the peculiar name of the medicines and instruments which stands out in the whole sentence compared to a dataset from restaurant with commonly used words such as food, waiter, service,etc

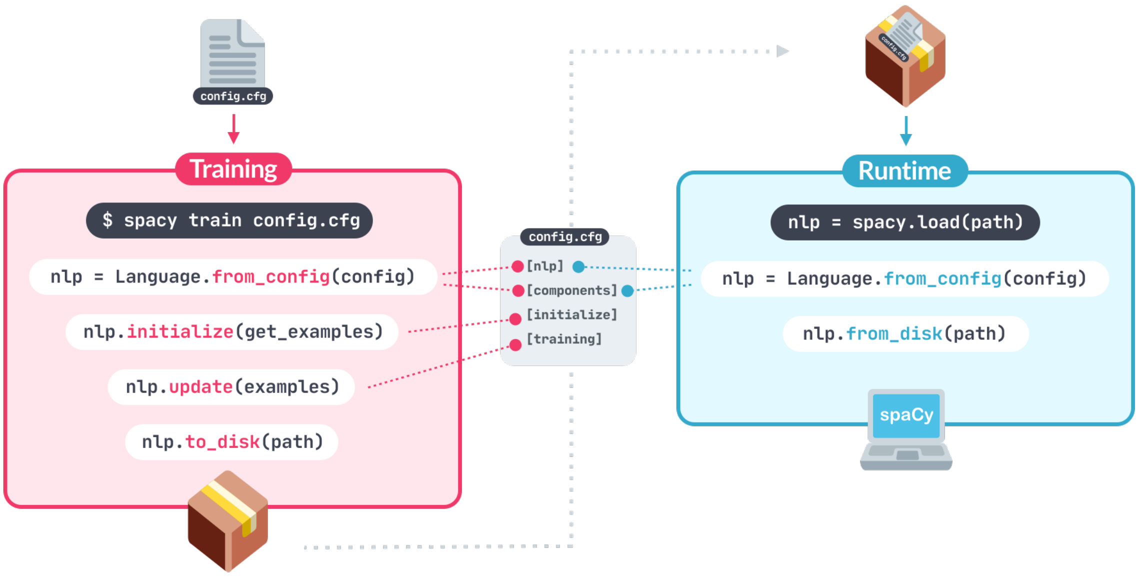


Figure 3.1: spaCy Model[11]

The bias surrounding the English language and food dishes also plays a big role. For example, popular western dishes like pizza, burger, pastries have higher frequency in dataset as opposed to Indian Dishes likes Naan, Paneer, Roti, etc.

Due to the aforementioned difficulties and low accuracy the model was discarded and we moved on with the Bidirectional LSTM model based on Recurrent Neural Network(RNN).

3.1.2 Bidirectional LSTM Model

LSTM stands for Long Short Term Memory. LSTM's are Recurrent Neural Networks (RNN).

- Recurrent Neural Networks:-
  RNNs are a type of artificial neural network that works with time series or sequence data. Artificial Neural Networks can only handle data points which do not share a context. They fail to take into account the context and relation between words. When dealing with sequential data, where one data point is dependent on the previous one, the neural network's design must be altered to account for these dependencies. RNNs have a memory notion that allows them to save the states or information from previous inputs in order to create the current out and pass it as input for the next iteration.

  The information flow in a feed-forward neural network is unidirectional. The data never passes through the same neuron twice. Feed-forward neural networks have no memory of the information they receive and are poor predictors of future events.

The information in an RNN runs in a loop. It evaluates the current input as well as results from previous inputs before arrving to a result.

As a result, there are two inputs to an RNN: the present and the result from the previous iteration. This is significant as it helps predict what will happen next, which is why an RNN can perform tasks that other algorithms cannot.

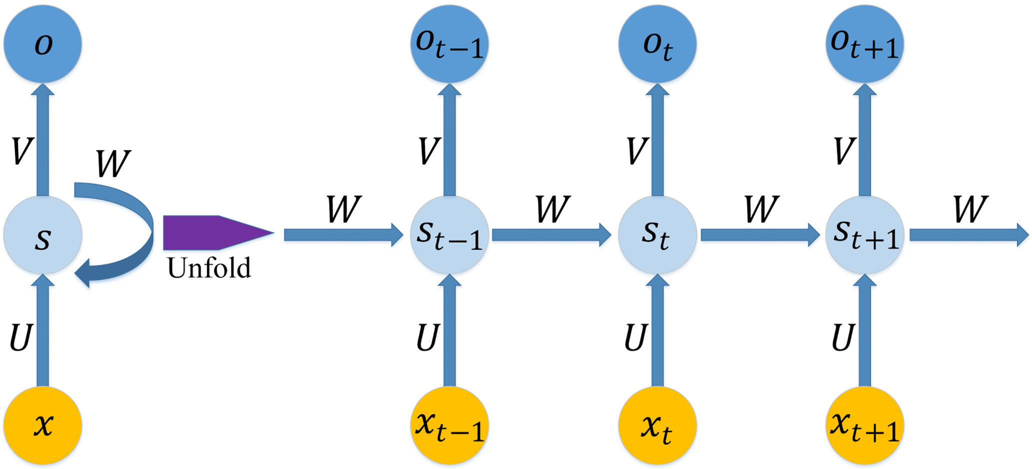


Figure 3.2: Recurrent Neural Network and the Unfolding Architecture[10]

But despite this, RNN suffers from short term memory. RNN are not able to remember information for a long time. The reason is the issue of the vanishing and exploding gradients. This is because the multiplication of a number in the zero-to-one range with another number in the same range results in an even smaller number than the initial two.

Some use cases of RNNs are:-

- Named Entity Recognition (Pharmacies)
- Langauge Translation (Google Translate)
- Sentiment Analysis (Yelp)
- Next words prediction (Gmail)

To overcome this problem LSTM i.e. Long Short Term Memory RNNs were designed.

- Long Short Term Memory RNN:-
  LSTMs can learn long-term dependencies. They are specifically developed to prevent the problem of short-term memory.

  The linked structure of LSTMs is similar to that of RNNs, but the repeating module is different. There are four neural network layers, all interacting in a defined way.

  The LSTM can delete or add information using gates. Gates are a mechanism to selectively allow information to pass through. A sigmoid neural net layer produces integers ranging from zero to one, indicating how much of each component should be allowed to pass. Zero means None and One mean All.

  There are three gates, input gate, output gate and forget gate.

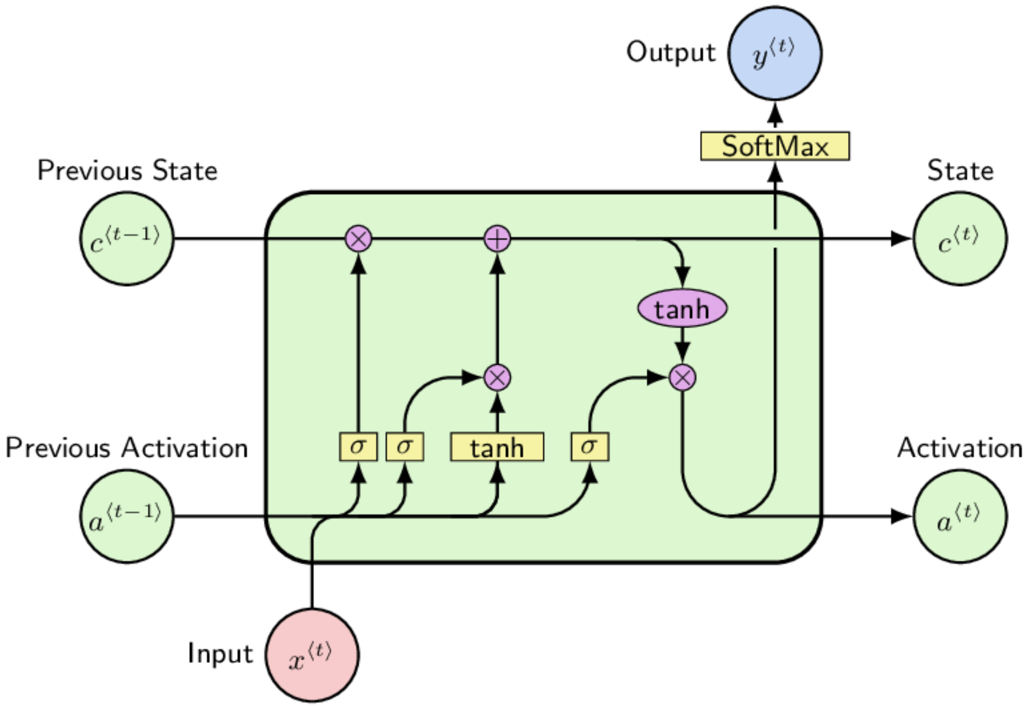


Figure 3.3: LSTM Cell Architecture[16]

A better version of LSTM is bi-directional LSTM

- Bi-directional Long Short Term Memory RNN:- Bidirectional LSTMs train two LSTMs instead of one. The first is based on the original input sequence, and the second is based on a reversed replica of the original input sequence. This can give the network more context and help it learn the problem faster and more thoroughly.

  Information flows in both directions, forward (past to future) and backward (future to past).

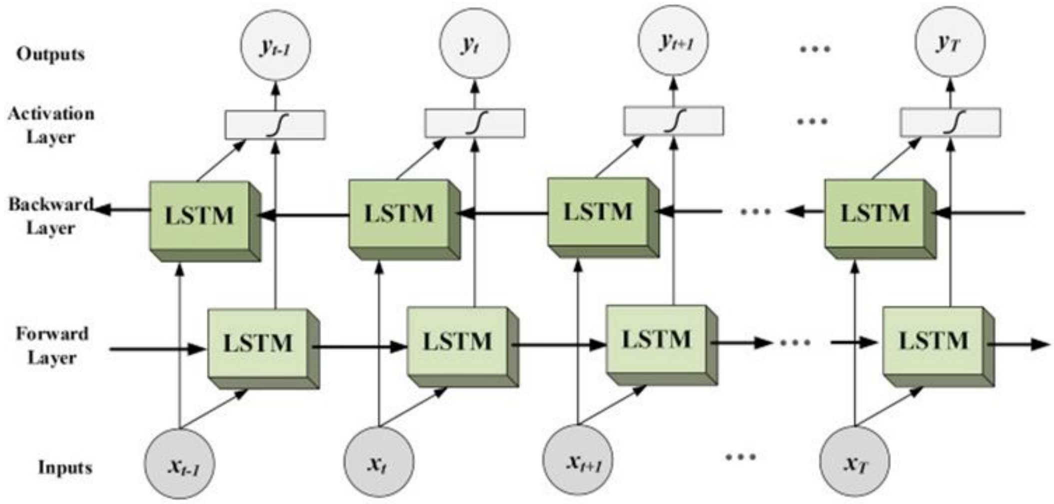


Figure 3.4: Flow of Bi-directional LSTM[8]

The model created with the help of Bi-directional LSTM faced the problem of class imbalance due to bad dataset. We tried oversampling but it was not of much use. The accuracy was no were near the spaCy method hence we tried the next method, using pre-trained BERT model.

3.1.3 BERT Model

BERT (Bidirectional Encoder Representations from Transformers) is a pre-trained language model that considers a word's context from both the left and right sides at the same time. It is different from previous model as it uses attention mechanism[33] which is different from the sequence to sequence framework used by LSTMs. The left and right pre training of BERT is achieved using Masked Model Language(MLM). MLM masks a random word in the sentence and then tries to predict that word by using the context from both right and left using transformers[7].

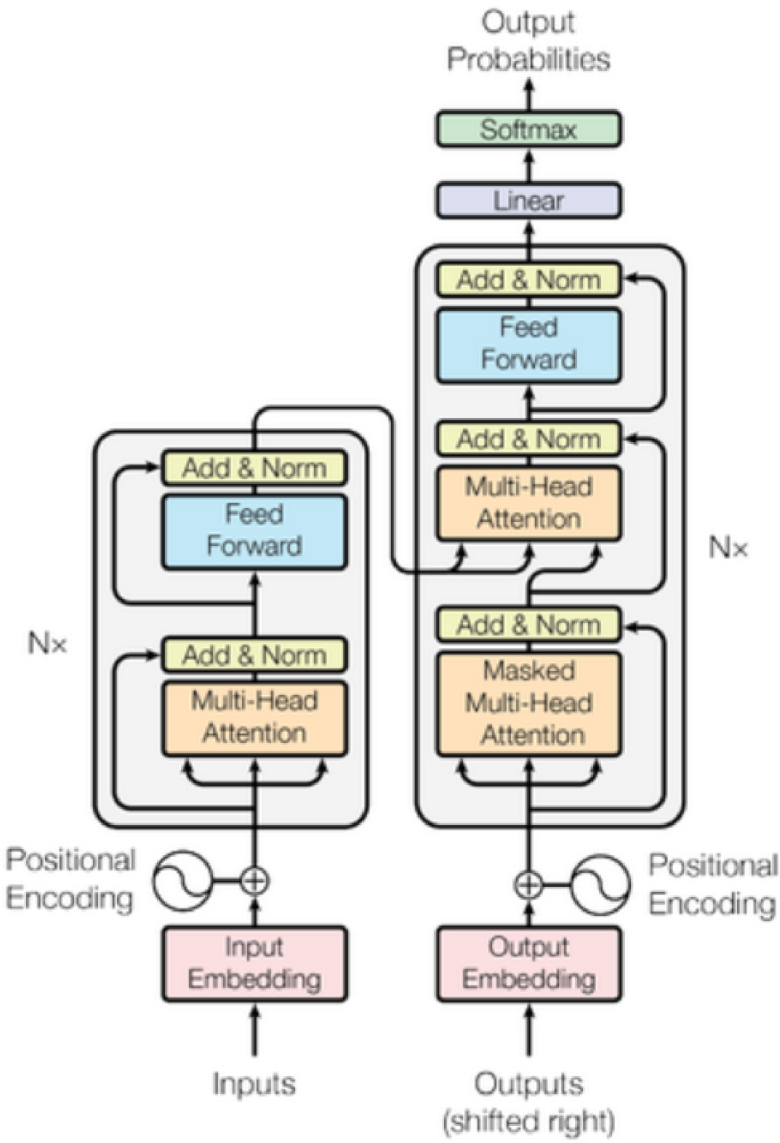


Figure 3.5: Transformer Components[13]

Transformers are built around attention mechanics rather than RNNs, which determine which sequences are important in each computational step. The encoder not only converts the input to a higher-dimensional space vector, but it also feeds the decoder with the necessary keywords. As a result, the decoder improves since it now has more information about which sequences are significant and which keywords provide context for the sentence[7].

To understand the relationship between two text sentences, Next Sentence Prediction is applied. BERT has been pre-trained to anticipate whether or not two sentences have a relationship.

There are two variants of BERT available:

- BERT Base; having 12 layers of transformer blocks, 12 attention heads and 110 million parameters

- BERT Large; having 24 layers of transformer blocks, 16 attention heads and 340 million parameters

We have used the BERT base model so that we can train it faster as compared to the large variant.

3.2 Word Embedding

Word embedding allow us to utilise a compact, dense representation with similar encoding for similar words. We don't have to define this encoding manually, which is a big plus. An embedding is a dense vector of floating-point values with a length determined by a parameter. The values for embedding can be attained by training. When working with huge datasets, it's not uncommon to see word embedding of up to 1024 dimensions. Higher dimensional embedding can capture finer-grained word associations[30], but they require more data to learn.

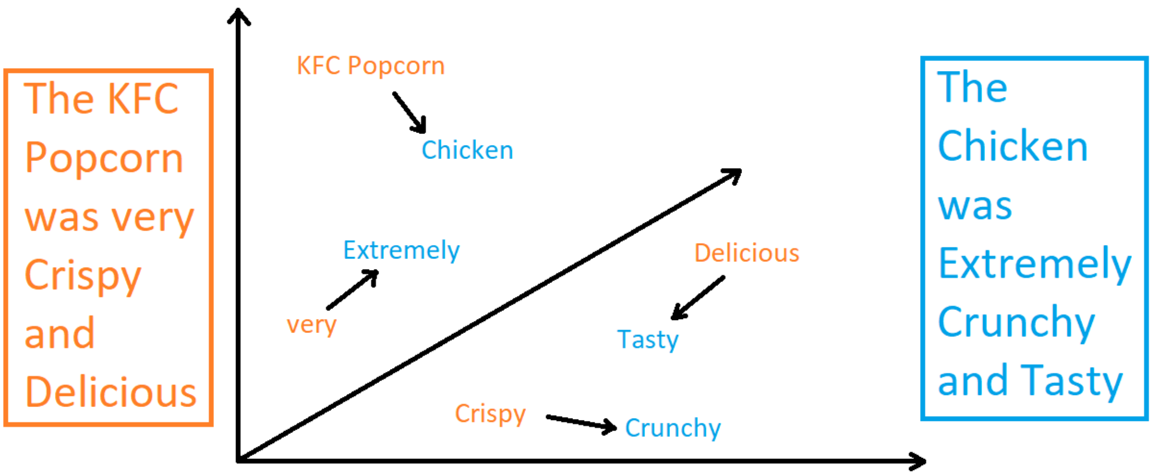


Figure 3.6: Word Embedding Chart

There are three ways to learn word embedding:-

- **Embedding Layer**
  For want of a better term, an embedding layer is a word embedding trained in conjunction with a neural network model on a specific natural language processing job, such as language modelling or document categorization.

- **Word2Vec**
  The study includes analysing the taught vectors and experimenting with vector math on word representations. For example, taking the ”man” out of ”King” and adding ”woman” yields ”Queen,” which captures the connection ”king is to man what queen is to woman”[24].
  There are two models that use the word2vec approach to for learning word embedding:-

  – Continuous Bag of Words
  – Continuous Skip Gram Model

- **GloVe**
  A more effective way of learning word representations is the GloVe(Global Vectors for word representations) algorithm, which is an extension of the word2vec method, developed at Stanford. GloVe tries to combine two things, one is global statistics of matrix factorization and the other is contextual learning in word2vec[26].

For our purpose, we have utilised Continuous Bag of Words, which employs the word2vec technique, because it is capable of learning high-quality word embedding quickly.

3.3 Sentiment Classification Model

We tried two models in Sentiment Classification:-

- Logistic Regression
- Bi-directional LSTM

3.3.1 Logistic Regression

The dataset used was Restaurant Reviews taken from SemEval2016. Some preprocessing was done on the dataset like removal of punctuation marks, stopwords and lemmatization. We used TF-IDF(Term Frequency-Inverse Document Frequency) vectorizer to convert the sentences to vectors. The formula for the same is

$$W(i,j) = tf(i,j) * log(N/df(i))$$

W(i,j) is the weight of word i in document j
tf(i,j) is the frequency of word i in document j
df(i) is the total number of documents containing word i
N is the total documents

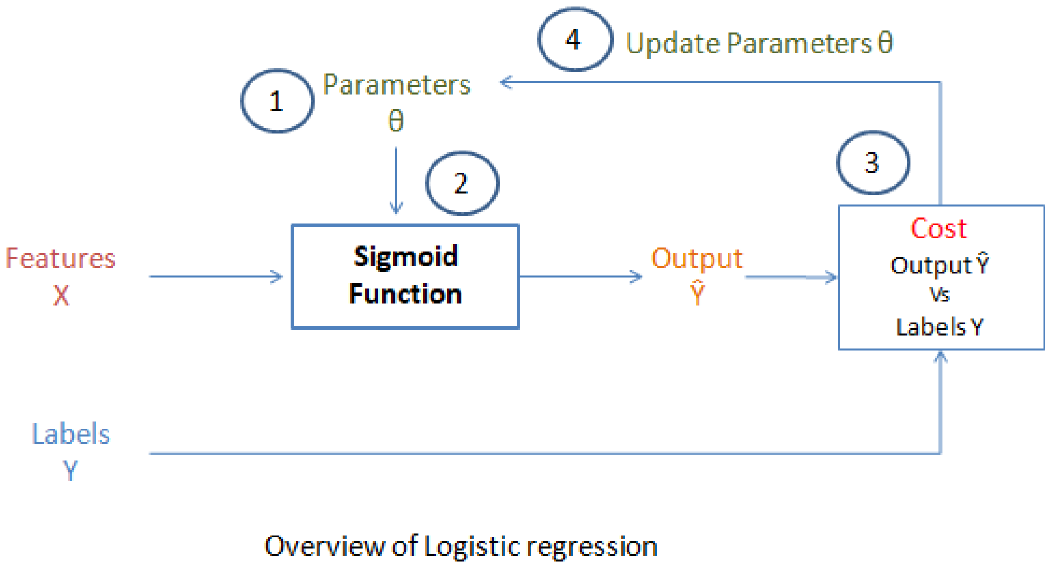


Figure 3.7: Overview of Logistic Regression[12]

Then we trained a logistic regression classifier to predict the sentiment polarity for a given piece of text[29]. Logistic regression uses a logistic function defined below to model a output variable which is either 1 or 0. Logistic regression uses a loss function known as Maximum Likelihood Estimation (MLE) which is a conditional probability. Depending on the probability, the predictions are classified into class 0 or 1. If the probability is greater than 0.5, it is predicted as class 0 else 1.

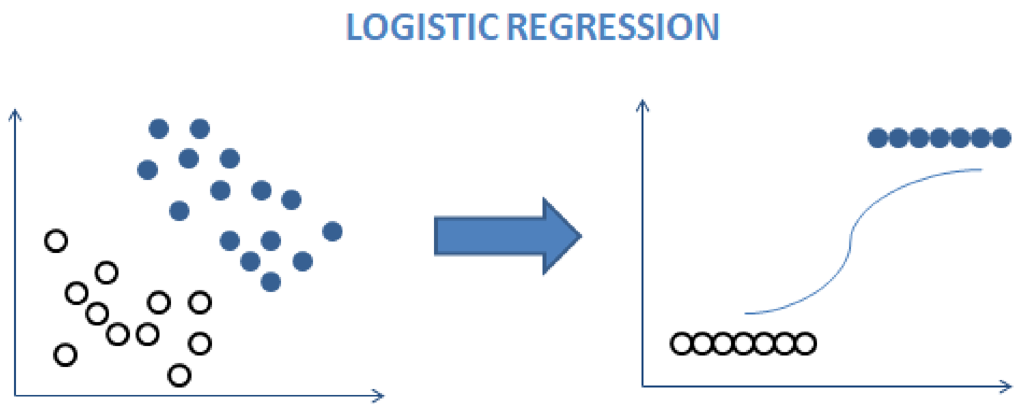


Figure 3.8: Logistic Regression Result[12]

3.3.2 Bi-directional LSTM

The dataset used was IMDB reviews dataset from kaggle website. The dataset was cleaned using gensim library with removed punctuation and did lemmatization. Then stops words were removed and special care was taken to not remove negations like do not, will not, never, etc.

Bi-directional Long Short Term Memory RNN:- Bidirectional LSTMs train two LSTMs instead of one. The first is based on the original input sequence, and the second is based on a reversed replica of the original input sequence. This can give the network more context and help it learn the problem faster and more thoroughly. Information flows in both directions, forward (past to future) and backward (future to past).

The model had following layers:-

- Embedding Layer
- Bidirectional Layer with 200 output Units
- Bidirectional Layer with 256 output Units
- Bidirectional Layer with 64 output Units
- Dense Layer with 1 output neuron and sigmoid activation function

- **Embedding Layer**

  We created out own word embedding using CBOW method. But the word vectors did not seem to provide a good accuracy. Hence we used google pre trained word embedding which have a dimension of 300. Sentences of 64 words were used, and

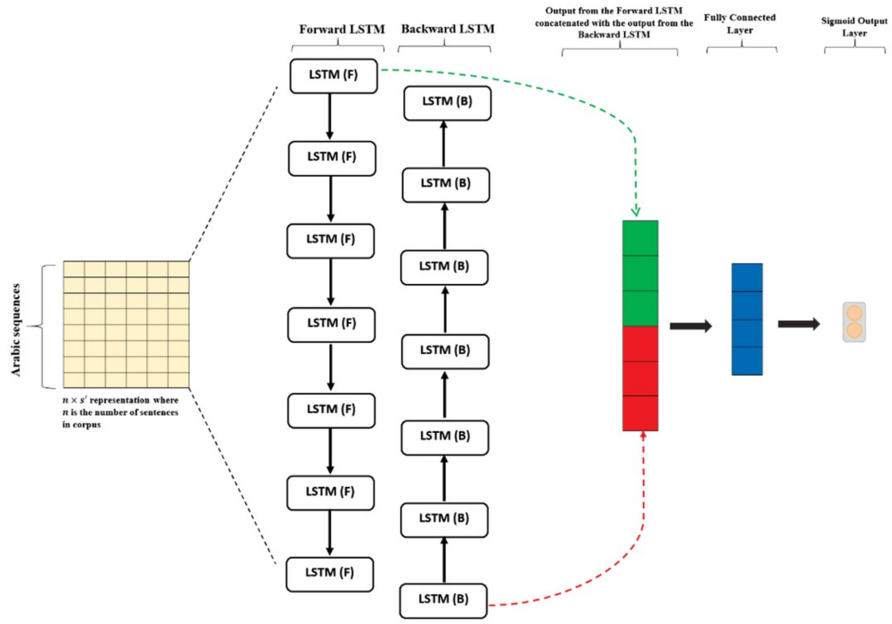


Figure 3.9: Bi-directional LSTM Model[14]

hence both padding and truncating was used. The reason behind 64 words was that we were going to use Constituency parsing to break down sentences into sub-phrases which were not expected to be of greater length.

- **Bidirectional Layer with 200 output Units**

  A total of 3 combinations with 200, 450 and 600 outputs were tried matched with permutations of other parameters. The best output units for this layer came out to be 200.

- **Bidirectional Layer with 256 output Units**

  A total of 3 combinations with 64, 128 and 256 outputs were tried matched with permutations of other parameters. The best output units for this layer came out to be 256.

- **Bidirectional Layer with 64 output Units**

  A total of 3 combinations with 64, 128 and 256 outputs were tried matched with permutations of other parameters. The best output units for this layer came out to be 64.

- **Dense Layer with 1 output neuron and sigmoid activation function**

  All the output will be passed on to a dense layer with one neuron. The activation function is sigmoid since it is a multi-label classification with many possible answers. Hence sigmoid will give range between 0 and 1. Here tending to 0 means negative review and tending to 1 means positive review. Tending to in between, i.e. near 0.5 can be considered as neutral review but it is a bias since the input had values only in binary

3.4 Constituency Parsing

Constituency Parsing is the process of examining sentences by breaking them down into constituents (sub-sentences).These constituents can belong to noun phrase, verb phrase, cellular sentence, etc.

Constituency parsing is done with the help of a structure called parse tree. A parse tree, is an ordered, rooted tree that describes a string's syntactic structure according to some context-free grammar. The parse tree used here is called constituency parse tree. The constituency parse tree is based on context-free grammars' formalism. The sentence is broken into constituents in this form of tree, which are sub-sentences that belong to a given grammar category.

We used StanFord CoreNLP's Stanza library for parsing. The output is in the form of tree structure. The three processors in the stanza pipeline that were used were tokenizer, POS and constituency. Others alternate options are provided by spaCy and also NLTK. But for our purpose Stanza worked out best.

The need behind this arises from the fact that a single sentence can have multiple aspects and their corresponding sentiments or even multiple aspects with one common sentiment. Since the above two models, both work on full sentences, the accuracy can get drastically affected if we are not able to pin point specific aspect and corresponding sentiment pairs.

Here a sentence will be broken down into its sub phrases. Hence a sentence can have one aspect and one corresponding sentiment, more than one aspect and one common sentiment for all of them or no aspect or no sentiment.

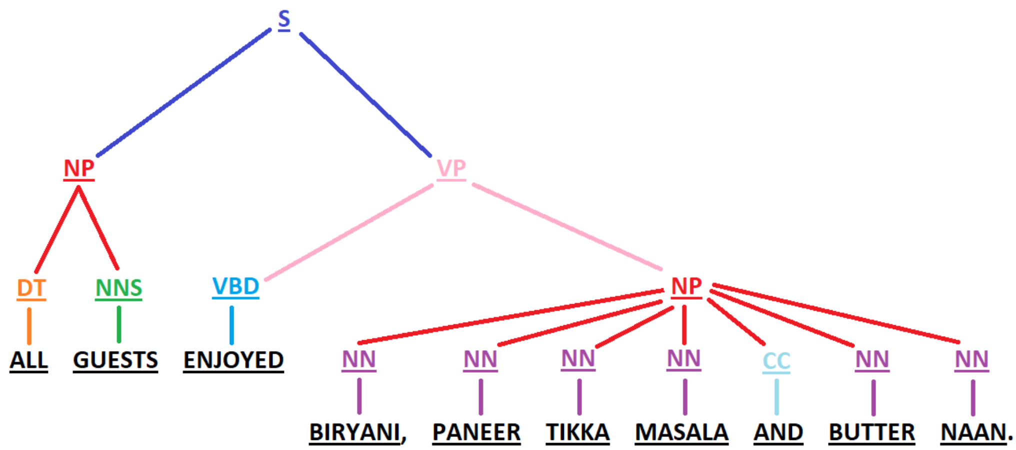


Figure 3.10: Constituency Parsing

Hence constituency parsing does the task of sub dividing sentences into sub sentences which are then fed to Aspect Category Detection Model and Sentiment Classifier.

3.5 Speech to Text Conversion

Speech to Text conversion is a sub topic under Natural Language Processing or Natural Linguistic Computation. When a person speak he creates disturbance in the surrounding air which gets picked up by a microphone. This is then passed Analog-to-Digital Converter(ADC) to obtain a digital waveform of the audio signal. Afterwards, the system takes precise measurement of sample at frequent intervals. All the extra noise is removed and changes the amplitude to bring the whole audio into one constant volume level. Other changes such as normalizing the speed of speech is performed. After these basic steps, the signal is divided into very tiny parts of around hunderdths of a second. The system then compares each part in context with parts surrounding it with known 40 phonemes in English language to recognize the exact sound and accordingly form words.

For the purpose of speech to text conversion we have used the SpeechRecognition library in python. This library allows us to perform above steps easily using python.

Chapter 4

Challenges

The following challenges were faced in the respective models:-

4.1 Aspect Category Detection

The challenges faced in Aspect Category Detection were

- Lack of proper dataset including absence of BIO tagging and improper data points.
- Class Imbalance in dataset due to the overloading of 'Other' category Aspect.
- spaCy's bias to only detect entities of some particular filed. Failure of Custom named Entity Recognition.
- Inability of Bi-directional LSTMs to give better results as they only take in information in one context.
- Heavy load on machine in fine tuning BERT model due to its complexity and time of training.
- Difficulty in selecting proper parameters and running more epochs due to limitations on the machine.
- Lack of flexibility provided by simple-transformers library to alter the architecture of BERT.
- Multi-threading error while saving BERT model in windows as it is relatively new model. The test script goes on calling itself in an infinite Loop. The matter was resolved by switching into Ubuntu.

4.2 Sentiment Analysis on Sentence

The challenges faced in Sentiment Classifier were:-

- Blind prepossessing that results in altering the original words which may change meaning of a sentence.
- Removal of stop words which also removed negative words like not, never, etc. Thus altering the meaning of sentence and providing with wrong results.
- Bad accuracy using Logistic Regression.
- Relatively bad accuracy using LSTM as compared to Bi-directional LSTM.
- Not much improvement despite using out own word vectors. Though the accuracy improved after using pre-trained Google word embedding.
- Single Context information taken into account by Bi-directional LSTM as compared to pre-trained BERT model for sentiment Analysis.
- End result in the form of range between zero to one, to make the model more versatile. But the input was only in binary and hence lack of accuracy in detecting neutral reviews.
- Inaccuracy in labeling few instances of 'good' with a positive sentiment.

4.3 Constituency Parsing

The challenges faced in Constituency parsing were:-

- Selecting proper library for generating parse tree. We went with Stanza library by Stanford CoreNLP group.

4.4 Speech to text Recognition

The challenges faced in speech to text were:-

- We first tried to take help of MATLAB for speech to text to conversion but the problem here was that it used third party APIs like Google Speech API, Microsoft Azure Speech API etc., for the purpose of speech to text transcription. To avail the services of these APIs, we had to pay a subscription fee, for which we were hesitant. So we then switched to python for speech to text conversion.

Chapter 5

Results and Discussions

5.1 Aspect Category Detection

5.1.1 spaCY Model

Results for Aspect Category Detection using the spaCY Model are shown below:-

Figure 5.1: spaCY Model Result

Findings: Low accuracy of 77% and was not able to detect all aspects. As seen from the first result, any specific food names and ”loud and noisy family” being part of Ambience#General was not recognized.

5.1.2 Bi-directional LSTM Model

Results for Aspect Category Detection using the Bi-directional LSTM Model are shown below :-

```
('You must install pydot (`pip install pydot`) and install graphviz (see instructions at https://graphviz.gitlab.io/downlo
ad/) ', 'for plot_model/model_to_dot to work.')
11/11 [==============================] - 18s 796ms/step - loss: 0.7549 - accuracy: 0.8902 - val_loss: 0.2732 - val_accurac
y: 0.9831
11/11 [==============================] - 7s 638ms/step - loss: 0.2733 - accuracy: 0.9831 - val_loss: 0.2732 - val_accurac
y: 0.9831
11/11 [==============================] - 7s 601ms/step - loss: 0.2732 - accuracy: 0.9831 - val_loss: 0.2732 - val_accurac
y: 0.9831
```

Figure 5.2: Bi-Directional LSTM Model Result

Findings: Very high accuracy of 98% but suffers from class imbalance. In the dataset, the tag ‘O’ was overloaded. Hence every tag was getting identified as ‘O’ only.

5.1.3 BERT Model

Results for Aspect Category Detection using the BERT Model are shown below:-

```
args = NERArgs()
args.num_train_epochs = 16
args.learning_rate = 1.5*1e-4
args.overwrite_output_dir = True
args.train_batch_size = 128
args.eval_batch_size = 128
#args.fp16 = True
```

```
result
```

```
{'eval_loss': 0.5261560860607359,
 'precision': 0.5656192236598891,
 'recall': 0.6035502958579881,
 'f1_score': 0.583969465648855}
```

Figure 5.3: Results and Parameters of the BERT Model with most accurate results

```
s8, model_output8 = m.predict(["The pizza was bad, the waiter was rude. The location was disgusting. The evironment was smelly"])
```

```
  0%|          | 0/1 [00:00<?, ?it/s]

Running Prediction:   0%|          | 0/1 [00:00<?, ?it/s]
```

```
prediction8
```

```
[[{'The': 'O'},
  {'pizza': 'FOOD#QUALITY'},
  {'was': 'O'},
  {'bad,': 'O'},
  {'the': 'O'},
  {'waiter': 'SERVICE#GENERAL'},
  {'was': 'O'},
  {'rude.': 'O'},
  {'The': 'O'},
  {'location': 'LOCATION#GENERAL'},
  {'was': 'O'},
  {'disgusting.': 'O'},
  {'The': 'O'},
  {'evironment': 'AMBIENCE#GENERAL'},
  {'was': 'O'},
  {'smelly': 'O'}]]
```

Figure 5.4: Results for an example

Findings: High accuracy and ability to pinpoint all aspects. Used in final model.

5.2 Constituency Parsing

Result of using Stanza library by Stanford NLP Group is shown below:-

```
def sentence_to_sub_phrases(root, roots):
    sub_sent = False
    for i in range(len(root.children)):
        if sentence_to_sub_phrases(root.children[i], roots):
            sub_sent = True
    if sub_sent == False and (root.label == 'S' or root.label == 'SINV'):
        roots.append(root)
        return True
    if sub_sent == True:
        return True
    return False
```

```
doc = nlp('Service was good but food was tasteless and bad')

for sentence in doc.sentences:
    print(sentence.constituency)
```

```
(ROOT (S (S (NP (NN Service)) (VP (VBD was) (ADJP (JJ good)))) (CC but) (S (NP (NN food)) (VP (VBD was) (ADJP (JJ ta
steless) (CC and) (JJ bad))))))
```

Figure 5.5: Constituency Parsing Result

5.3 Sentiment Polarity Classifier

5.3.1 Logistic Regression Model

Result for Sentiment Polarity Classifier using the Logistic Regression Model are shown below:-

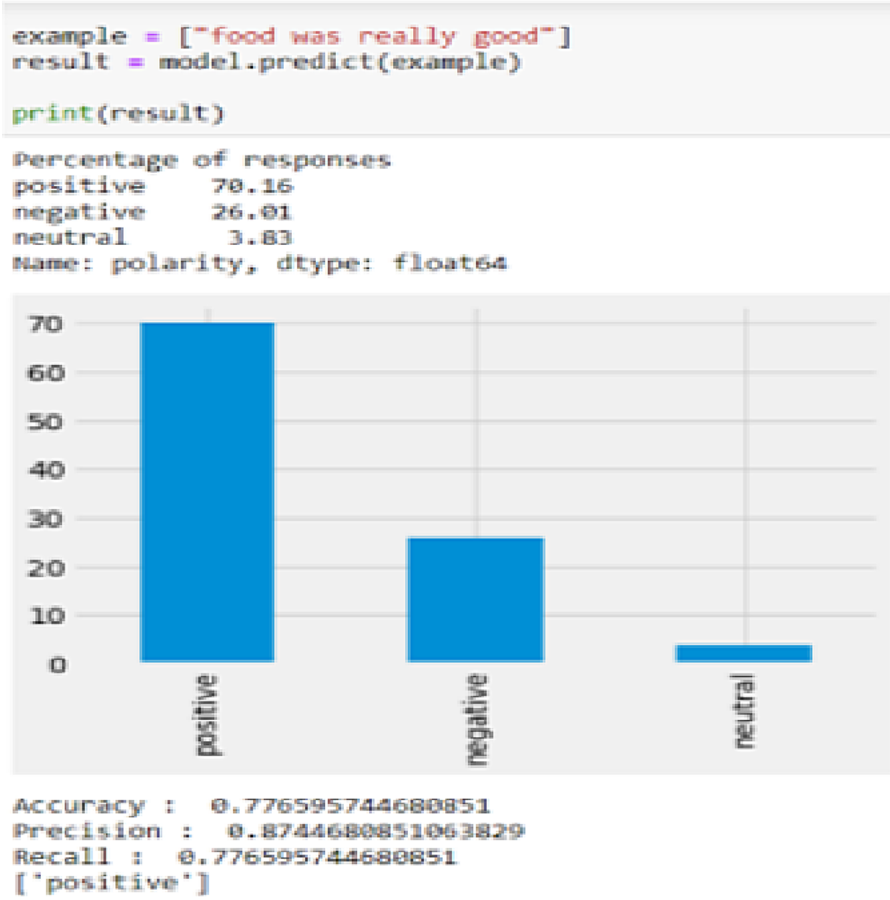


Figure 5.6: Logistic Regression Model

Findings: Low accuracy due to not being able to correctly classify any positive sentences with word ”not” in it. e.g. ”not bad”.

5.3.2 Bi-directional LSTM Model

Results for the Sentiment Polarity Classifier using the Bi-directional LSTM Model are shown below:-

```
score = model1_2.evaluate(x_test1, y_test1, verbose = 1)

print('Test loss:', score[0])
print('Test accuracy:', score[1])
```

```
616/616 [==============================] - 2s 3ms/step - loss: 0.1911 - accuracy: 0.9202
Test loss: 0.19109366834163666
Test accuracy: 0.9202070832252502
```

Figure 5.7: Accuracy Results for Bi-LSTM Model

```
: txt = ["Overall, not a good experience"]
  sequences = gensim.utils.simple_preprocess(txt[0])
  words = [word for word in sequences if word not in modified_stop_words and word != 'br']
  print(words)
  sentences = [[vocab_to_int[w] for w in words]]
  sentences

  data_temp = pad_sequences(sentences, truncating='post', padding = 'post', maxlen=64)
  data_temp

  model1_2.predict(data_temp)
```

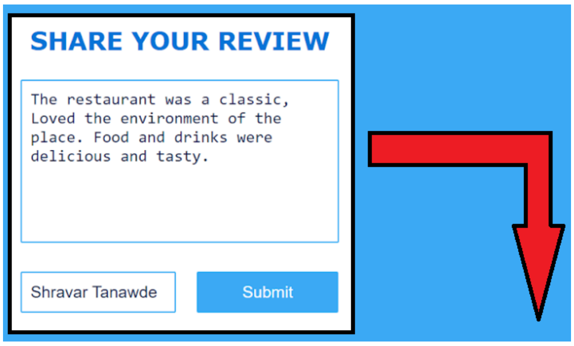
```
  ['overall', 'not', 'good', 'experience']

: array([[0.00257126]], dtype=float32)
```

Figure 5.8: Results for an example

Findings: High accuracy of 92.02% and able to correctly classify sentiments of an aspect. Used in final model.

5.4 Aspect Based Sentiment Analysis Model

Result of Whole model which includes Aspect Category Detection Model(BERT Model) as well as Sentiment Polarity Model(Bi-directional LSTM Model):-




```
para = 'The resturant was a classic. Loved the environment of the place. The food and drinks were delicious and tasty
input_feedback(para)
```

```
The resturant was a classic. Loved the environment of the place. The food and drinks were delicious and tasty. The s
ervice was not great, needs improvement on that part. Food was great but service was bad. But my overall experience
was good


The resturant was a classic.
HBox(children=(HTML(value=''), FloatProgress(value=0.0, max=1.0), HTML(value='')))


HBox(children=(HTML(value='Running Prediction'), FloatProgress(value=0.0, max=1.0), HTML(value='')))

The resturant was a classic . --> resturant AMBIENCE#GENERAL 0.9993133


Loved the environment of the place.
HBox(children=(HTML(value=''), FloatProgress(value=0.0, max=1.0), HTML(value='')))


HBox(children=(HTML(value='Running Prediction'), FloatProgress(value=0.0, max=1.0), HTML(value='')))

Loved the environment of the place . --> environment AMBIENCE#GENERAL place AMBIENCE#GENERAL 0.9994588


The food and drinks were delicious and tasty.
HBox(children=(HTML(value=''), FloatProgress(value=0.0, max=1.0), HTML(value='')))


HBox(children=(HTML(value='Running Prediction'), FloatProgress(value=0.0, max=1.0), HTML(value='')))

The food and drinks were delicious and tasty . --> food FOOD#QUALITY drinks DRINKS#QUALITY 0.99952674
```

Figure 5.9: Results for the whole model

Chapter 6

Conclusions

6.1 Project Summary

During this Project, we tried to implement different machine learning methodologies to apply NLP on text to obtain desired results. Our objectives were:

1. Applying Aspect Category Detection model of a large body of text and successfully recognize all the aspects mentioned in that text.
2. Applying Sentiment Polarity Classifier model on the previously obtained aspects of text and classify each aspect as negative, neutral or positive.
3. Combining best of both models into one large model called Aspect Based Sentiment Analysis model.
4. Applying speech-to-text as a complement to ABSA model.

For fulfilling the above goals we implemented various methodologies for both models namely,

- Aspect Category detection
    1. spaCY Model
    2. Bi-directional LTSM Model
    3. BERT Model
- Sentiment Polarity Classifier
    1. Logistic Regression Model
    2. Bi-directional LTSM Model

After testing all the aforementioned methods we were able to deduce the pros and cons of every method and used the best method for each model. Our findings have been recorded in the following sections.

6.2 Conclusion from Aspect Category Detection

Out of the three methodologies we used for this purpose we had to use BERT model. BERT Model gave the most desired result out of all three. The conclusion from all three methods are as follows:

spaCY Model

spaCY model was the first method implemented to create this model. spaCY is an easy to use model because a lot of NLP functionality is already available in it. Upon training the model, results were not satisfactory. This method gave pretty low accuracy. The main issue was inability of this model to recognize all the aspects present in the given text. From the result main culprits behind this issue is whenever a specific name is used. Things like "Chow Fun" and "Pork Shu Mai" was supposed to be classified as Food#Quality but the model failed to do so. Another instances of this issue were that model was not able to classify "Loud and rude family" as Ambience#Quality and problems with waiter as Service#General. Hence, as the results were not desirable, this method was scraped.

Bi-directional LSTM Model

BiLSTM model was the second model we tested and this one showed very promising result. We were able to get a theoretical accuracy of over 98% consistently. The issue with this method, however, was the class imbalance error. Class imbalance error occurs when the dataset used for training has a very large majority of one class of sample compared to other classes. In our dataset most of the samples were tagged as 'O' for NULL aspect. Due to this 'O' class got overloaded and this method was not able to accurately apply tags other than 'O'. Hence this method had to scraped due to our inability of successfully solving the issue of class imbalance.

BERT Model

After obtaining not so desirable result from first 2 methods we tested BERT Model. Even though this method also suffered from class imbalance issue, we were able to quickly resolve it. Only task required for using this method was to preprocess the dataset and hyper-parameters training. This gave us a model with high accuracy and was able to pin-point all the aspects present in the text. Satisfied with result we decided to use this method for the final model.

6.3 Conclusion from Sentiment Polarity Classifier

Similar to Aspect Detector Model we tested 2 methods for this model. Bi-directional LSTM method was the most useful of the two and hence it was used in the final model. Following is conclusion of both the method that was tested:

Logistic Regression Model

Logistic Regression method was our first choice. This method showed promise but had issued with statement using the word ”NOT” or any other word with similar effect. So this method classifies statement according to ”value ” assigned to each word in a sentence. So words such as ”Happy”, ”Tasty”, ”Clean”, etc is given high value and so most sentence with these words are classified as positive sentiment. Similarly, words such as ”Bad”, ”Poor”, ”Dirty”, etc corresponds to negative sentiment. But due to this, statements such as ”Food was NOT BAD” gets wrongly classified as negative sentiment. This issue brought down the accuracy of this method to 77% and hence this was scraped.

Bi-directional LSTM Model

This method did not have any downsides to it. It gave a very accuracy of 92.02% and did not suffer from the issue that previous method had. Hence this method was chosen to be used in the final method.

With this the final ABSA model was a combination of BERT Model and Bi-directional LSTM model. There could be much better methods than the one we decided to use and hence there is still possibility of being able to improve the model in future.

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